

Planting Date Influence on Yield of Drought-Tolerant Maize with Different Maturity Classifications

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ABSTRACT

Planting date and selection of an appropriate hybrid are critical components in optimizing maize (*Zea mays* L.) yield. The objective of this study was to determine the influence of planting date on yield and yield components of DroughtGard maize hybrids with different maturity classifications as compared to a conventional maize hybrid. Planting date within year had no influence on maize yields of 10.8 Mg ha⁻¹ in 2013 and 13.6 Mg ha⁻¹ in 2014. Delayed planting reduced 100-kernel weight by 4.1 to 6.0 g, and bulk density by 1.3 to 1.8 kg m³ in both years. Also, delayed planting reduced the number of ears m⁻² by 0.7 in 2014, but had no effect in 2013. Delayed planting decreased kernel depth in 2013 by 0.7 mm, but increased depth by 0.6 mm in 2014. Maize grain yield was correlated with all yield components, but variation of correlation magnitude occurred among yield components across planting dates. The number of ears m⁻² was relatively more important for early planted maize ($R = 0.58^{**}$ vs. $R = 0.10$) while the kernel weight ($R = 0.45^{**}$ vs. $R = 0.09$), kernel depth ($R = 0.48^{**}$ vs. $R = 0.23^{**}$), and bulk density ($R = 0.66^{**}$ vs. $R = 0.18$) were relatively more important for late-planted maize. No maize yield differences occurred between DroughtGard and conventional hybrids with similar maturity. Planting date had little influence on grain yields for maize following soybean [*Glycine max* (L.) Merr.] due to grain yield component compensation effects.

Core Ideas

- Planting date had no influence on maize grain yield.
- Yield components varied across planting dates.
- Late-maturity maize hybrids yielded more than early maturity maize hybrids.
- Maize grain yield was not influenced by drought tolerance type.

DURING RECENT DECADES U.S. producers have been planting maize earlier in the spring (Kucharik, 2008). Maize kernel germination is temperature dependent with optimal germination at 30°C, and minimal germination at 9°C (Warrington and Kanemasu, 1983). The minimal temperature must be present or exceeded for the first 48 h after planting to avoid imbibitional chilling injury (Creencia and Bramlage, 1971). Kucharik (2008) found that earlier planting has increased maize grain yield by 0.06 to 0.14 Mg ha⁻¹ d⁻¹ in states with relatively short-growing seasons including Iowa, Michigan, Minnesota, Nebraska, South Dakota, and Wisconsin, but not in the longer-growing season states of Kansas, Missouri, Illinois, Indiana, Ohio, and Kentucky. In Nebraska, Swanson and Wilhelm (1996) found that the highest grain yield was produced when maize was planted 10 and 12 May, and produced lower yields when planted earlier or later. Irmak and Djaman (2016) and Grassini et al. (2011) found increasing maize yield occurred in Nebraska with earlier planting under irrigation, but a variable and small yield difference under rainfed conditions which varied across years. In states neighboring Nebraska, Assefa et al. (2012) found that increased maize yield in Kansas by 0.25 Mg ha⁻¹ d⁻¹ with earlier planting, and Perez-Bidegain et al. (2007) found that planting date influenced maize yield in only 1 yr out of 3 in Iowa. Sindelar et al. (2010) found that early maize planting by 23 April increased maize yield by 10% in low-stress environments, decreased yield by 30% with early season stress, and increased yield by 60% when full-season stress was present. Minnesota research indicated no yield decrease occurred for a 2-wk delayed planting after 23 April until 7 May, while a 6 to 16% yield decrease for a 4-wk delay (Kaiser et al., 2016; Van Roekel and Coulter, 2011). Increased maize yield with early planting in states with short growing seasons, such as Nebraska, likely is associated with a longer growing season due to increased temperatures (Skaggs and Irmak, 2012) combined with planting of longer maturity maize hybrids (Kucharik, 2008; Lauer et al., 1999) with greater early season cold tolerance. Seifert et al. (2017) reported that crop rotation with soybean reduced the yield advantage of earlier planting of maize compared to maize following maize.

Several studies on the influence of planting date on maize grain yield components have been reported. Swanson and Wilhelm (1996) found that planting date had no influence on the number of ears m⁻² and little influence on kernel weight for rainfed areas, similar to results of Otegui et al. (1995) in Argentina who

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Abbreviations: CRM, comparative relative maturity; CspB, cold shock protein B; GDU, growing degree unit, base 10°C.

found that planting in late April and early May or planting in late May to early June reduced the number of kernels ear⁻¹ and kernels plant⁻¹. Van Roekel and Coulter (2011) and Kaiser et al. (2016) found a yield decrease with a 4-wk delay in planting in Minnesota was associated with a 7 to 9% decrease in kernel weight, similar to results of Zhou et al. (2017) in China. Lindsey and Thomison (2016) found that delayed planting reduced grain yield, kernel weight, number of kernels row⁻¹ and kernels plant⁻¹ while having no influence on the number of rows ear⁻¹ in Ohio.

Rainfed maize grain yields have increased at a rate of 50 to 120 kg ha⁻¹ yr⁻¹ in the United States during the last 75 yr (Assefa et al., 2012; Mason et al., 2008; Duvick, 2005). Selection of maize hybrids for native drought tolerance has been important in maize breeding programs, but recently there is increasing interest in use of biotechnology-driven drought-tolerant maize hybrids containing the bacterial cold shock protein B (CspB), hereafter called DroughtGard (Nemali et al., 2015). Under well-watered conditions, the DroughtGard maize hybrids containing the CspB transgene were reported to produce similar yields to conventional hybrids, but to produce 6% higher yield under water-limited conditions due to decreased leaf area thereby reducing water use during silking. Chang et al. (2014) found that DroughtGard and conventional maize hybrids produced similar yields in moderate- and high-yield environments, and had similar harvest index, root/shoot ratio, and silking leaf size as conventional hybrids. Kisekka et al. (2015) found similar yields in rainfed and with five irrigation levels, and that the DroughtGard hybrids had similar yield, crop water use, and water use efficiency to conventional hybrids, but produced lighter kernels. Nemali et al. (2015) reported that the DroughtGard maize hybrids produced similar number of ears m⁻² and kernel weights, and more kernels m⁻² than conventional maize hybrids under water-limited conditions, while Castiglioni et al. (2008) found increased number of ears m⁻² and number of kernels ear⁻¹ and similar kernel weights for DroughtGard and conventional maize hybrids.

Early maturity maize hybrids have lower yields than later-maturity hybrids due to lower leaf area indices, the consequent reduction in interception of photosynthetically active radiation, and decreased seasonal transpirational water use (Howell et al., 1998). Late-maturity maize hybrids have greater yield reduction due to delayed planting than early maturity hybrids (Andrade et al., 2005; Lauer et al., 1999). Dent maize hybrids in the United States were developed by hybridization of southern dent possessing short, girthy ears with northern flint germplasm possessing multiple ears plant⁻¹ and long, slender ears 8 to 10 rows ear⁻¹ composed of shallow kernels (Doebley et al., 1988). Milander et al. (2017) found

that early maturity maize hybrids had fewer rows ear⁻¹, and more kernels row⁻¹, and kernels ear⁻¹ than mid- and late-maturity maize hybrids over a range of plant populations.

Yield component compensation is common in maize with reduced early developing yield components resulting in increases of later occurring yield components (Milander et al., 2017; 2016). Early season growing conditions influence the number of ears m⁻² (or ears plant⁻¹) (Evans et al., 2003) and number of rows ear⁻¹ (Abendroth et al., 2011), while mid-season (mid-vegetative to mid-grain fill stages) growing conditions tend to influence the number of kernels ear⁻¹. Flowering (R2) is the growth stage most sensitive to water and heat stress that can reduce the number of kernels ear⁻¹ (Andrade et al., 2005; Abendroth et al., 2011; Westgate et al., 2004). Kernel abortion occurring during early grain fill can reduce kernel number. A stressful environment during grain fill can result in low kernel weights (Andrade et al., 2005; Abendroth et al., 2011; Novacek et al., 2013) while heavy kernel weight occurs with high irradiance and long grain-fill duration (Andrade et al., 2005; Novacek et al., 2014).

Although much research has been conducted on maize planting date, none have addressed use of modern DroughtGard hybrids, maturity classification differences, nor conducted a thorough yield component analysis as in this study. The objective of this research was to determine the influence of planting date on grain yield and yield components of four DroughtGard maize containing the CspB transgene with different maturity classifications as compared to a 109 CRM conventional maize hybrid.

MATERIALS AND METHODS

Field experiments were conducted in adjacent fields in 2013 and 2014 in rainfed environments with 30-yr April to October average of 456 mm at the University of Nebraska Agriculture Research and Development Center (ARDC) near Mead, NE. The predominant soil type was Tomek silt loam (fine, smectitic, mesic, Pachic Argiudoll) with 0 to 1% slope. The experiments were conducted in a randomized complete block design with split plot arrangement and three replications. Three planting dates of 27 April (soil temperature of 13.5°C; 153 growing degree units, base 10°C (GDU) after 1 April), 16 May (soil temperature of 21.8°C; 333 GDU after 1 April) and 3 June (20.9°C; 496 GDU after 1 April) in 2013 and 18 April (soil temperature of 8.9°C; 60 GDU after 1 April), 5 May (soil temperature 14.8°C; 155 GDU after 1 April) and 20 May (soil temperature 20.6°C; 258 GDU after 1 April) in 2014 nested within year was the whole plot. Maize hybrid with different maturity classification and drought tolerance traits was the split plot (Table 1). Plots were six rows wide (4.6 m wide) by 9.1 m long.

Table 1. Characteristics of maize hybrids used in this study (Monsanto and Channel, no date).

Hybrid	Drought classification	CRM†	GDU‡ to 50% silking	GDU to physiological maturity	Emergence score§	Seedling growth (vigor) score§	Drought tolerance score§
DKC 47-27	DroughtGard	97	1253	2375	2	2	2
DKC 50-57	DroughtGard	100	1269	2500	3	3	1
DKC 62-27	DroughtGard	112	1338	2800	3	3	2
Channel 214-00	DroughtGard	114	1350	2840	na¶	2	1
DKC 59-92	Conventional	109	1335	2725	2	2	2

† CRM = Comparative relative maturity.

‡ GDU = Growing degree unit, base temperature of 10°C.

§ 1 = excellent, ..., 9 = poor.

¶ na = not available.

Soybean was the previous crop. Production practices included conventional disk tillage before planting, and pre-emergent and post-emergent herbicides for weed control. Row spacing was 76 cm and the maize seeding rate was 65,000 seeds ha⁻¹. Early spring soil tests indicated pH of 6.0 to 6.4, organic matter concentrations of 28 g kg⁻¹, Mehlich-3 P concentration of 29 to 45 mL L⁻¹, and K concentrations of 323 to 412 mL L⁻¹, all greater than the sufficiency level for high-yield maize production in Nebraska (Shapiro et al., 2008). Fertilizer application was based on fall soil test for residual NO₃-N and organic matter, and expected grain yield of 10 Mg ha⁻¹. Thus based on expected yield, residual soil NO₃-N, soil organic matter concentration, and previous crop, 100 kg N ha⁻¹ was pre-plant applied in both years using the University of Nebraska N rate algorithm.

Early season plant populations were determined at the four-leaf developmental stage (Abendroth et al., 2011). Grain yield data was measured by mechanically harvesting the middle three rows of the plots at approximately 200 g kg⁻¹ water content. Grains were weighed, water content was measured, and yield for each plot was adjusted to a water content of 155 g kg⁻¹. Prior to harvest, the number of ears and plant population were counted to determine the number of ears m⁻². Six consecutive-ear samples were collected from each plot and used to measure the yield components. Primary and secondary yield components and related parameters measured were rows ear⁻¹, ear circumference, kernels ear⁻¹, kernels row⁻¹, kernel depth, ear length, and kernel weight. Rows ear⁻¹, kernels row⁻¹ and kernels ear⁻¹ were hand counted, and ear length and middle-of-the ear circumference were measured prior to hand shelling. Kernel depth was calculated from caliper measurements of ear and cob diameters. The number of kernels ear⁻¹ was hand counted and 100 kernels were randomly selected from each ear and used to determine the kernel weight. Water content was measured and 100-kernel weight was adjusted to a moisture content of 155 g kg⁻¹. Bulk density and grain water were determined using a Dickey-John GAC 2100 (Auburn, IL) moisture and bulk density meter.

Data were analyzed using PROC Mixed of SAS (SAS Institute, 2014). Analysis of variance was conducted with year, planting date nested within year, and hybrid main effects and their interactions which were considered fixed effects. Nesting of planting dates within year was necessary to avoid confounding since planting dates in 2014 were earlier and at lower soil temperatures than in 2013 (Littell et al., 2006). Delayed planting increased late-season plant populations by <2000 plants ha⁻¹ in 2013, and decreased late-season plant

populations by <6000 plants ha⁻¹ in 2014, but differences were not associated with grain yield ($R = -0.11$ and $R = -0.21$). Since plant population is often associated with yield components (Milander et al., 2016), a covariance analysis was conducted. To account for the variation in plant population, plant population was added as a covariant. The covariant analysis found only a change in the probability of year for the number of ear m⁻² (data not presented), and Pearson correlations among yield and yield components for adjusted means was almost the same as for non-adjusted means (data not presented). Based on these analyses, the original statistical analysis is presented rather than the covariance analysis. Pair-wise comparisons were used to assist with mean separation of planting date and hybrid responses. Pearson correlations were calculated to identify interrelationships among measured parameters.

RESULTS AND DISCUSSION

Seasonal Climatic Conditions

Total seasonal rainfall in 2013 was near the 30-yr average, with >50 mm more rainfall than monthly average in May, and >40 mm less than average in July and August (Table 2). In 2014, seasonal rainfall was >170 mm greater than the 30 yr average, with > 50 mm greater than the 30-yr monthly average in the months of June, August, and September and >50 mm less than average in July. In both years, average seasonal temperatures were near the 30-yr average. Monthly temperatures in 2013 were also similar to the 30-yr average, but in 2014 temperatures were 1.6°C higher than average in May and 2.6°C lower than average in July. Climatic conditions in both years were adequate for optimal pollination and high-yield rainfed maize production even with below-average rainfall in July in both years.

Grain Yield

Analysis of variance indicated that maize grain yield was influenced by year and maize hybrid, but not by planting date (Table 3). The lack of yield response to the planting dates contrasted with most published studies (Kaiser et al., 2016; Assefa et al., 2012; Van Roekel and Coulter, 2011; Swanson and Wilhelm, 1996), but is consistent with the Seifert et al. (2017) conclusion that soybean rotation reduces yield effects of planting date alteration in maize. However, similar responses were presented by Irmak and Djamin (2016) for rainfed maize in wet years and Grassini et al. (2011) for irrigated maize in Nebraska, and Perez-Bidegain et al. (2007) in Iowa. Average maize grain yield was 10.8 Mg ha⁻¹ in 2013, and 13.6 Mg ha⁻¹ in 2014 when almost 200 mm more rainfall was received between May and September than in 2013 (Table 2),

Table 2. Mean monthly rainfall and temperature for Mead, NE, in 2013 and 2014 during the maize growing season.

Month	Rainfall			Air temperature		
	2013	2014	30-yr average	2013	2014	30-yr average
	mm			°C		
Apr.	92	63	73	6.9	10.9	10.3
May	163	138	112	15.2	17.9	16.3
June	119	183	106	21.1	21.7	21.9
July	16	24	76	23.1	21.8	24.3
Aug.	46	166	89	23.6	23.3	23.0
Sep.	98	129	73	20.7	17.9	18.2
Oct.	98	33	58	11.0	11.8	11.2
May–Sept. Total/Average	442	640	456	20.7	20.5	20.7

Table 3. Mean squares and significance of grain yield and yield components influenced by year, planting date, and maize hybrid.

Source of variation	df	Grain yield	Ears m^{-2}	Rows ear^{-1}	Kernels ear^{-1}	Kernels row^{-1}	Kernel depth	Ear length	Ear circumference	Kernel weight	Bulk density
Year	1	173.06**	0.86	0.24	87,672**	319.15**	19.53	138.14**	19.41*	663.7**	8722**
Planting date(Year)	4	3.04	1.22**	1.76	5,584	1.85	2.19*	0.64	0.96	71.4**	1315**
Hybrid	4	21.06**	1.10**	28.25**	54,358**	8.54*	3.32**	4.24**	3.77**	70.3**	3162**
Year $\times$ Hybrid	4	2.16	1.32**	1.11*	910	3.52	1.44*	2.74*	1.87**	79.1**	871**
Planting date $\times$ Hybrid(Year)	16	1.20	0.25*	0.34	4,368**	11.78**	0.28	2.23**	0.21	1.88*	1

* Significant at the $P \leq 0.05$.** Significant at the $P \leq 0.01$.

including 64 mm more rainfall in June and 120 mm more rainfall in August. The late-maturity DroughtGard and check maize hybrids produced 1.8 to 2.1 $Mg\ ha^{-1}$ higher grain yield than by the early maturity DroughtGard maize hybrids (Table 4), consistent with results of lower grain yields for early maturity hybrids than late-maturity hybrids (Milander et al., 2017; Howell et al., 1998). No grain yield differences were found between DroughtGard and conventional hybrids with similar maturity as also found by Nemali et al. (2015).

Yield Components

Although planting date had no influence on maize grain yield in this study, yield compensation from alteration of yield components was found based on timing of environmental conditions with maize growth stages (Milander et al., 2017; 2016; Abendroth et al., 2011). Analysis of variance indicated complex interactions for Hybrid $\times$ Planting Date within Year for the number of ears m^{-2} , kernels ear^{-1} , kernels row^{-1} , and ear length (Table 3). In both years, planting dates between 27 April and 16 May resulted in production of a similar number of ears m^{-2} , but further delay slightly increased the number of ears m^{-2} in

2013 and decreased 0.7 ears m^{-2} in 2014 (Table 5). Change in the number of ears m^{-2} was associated with differences in early season ($R = 0.63-65^{**}$) and late-season ($R = 0.79^{**}-0.89^{**}$) maize plant populations. Otegui et al. (1995) found a decrease in the number of ears $plant^{-2}$ in Argentina with delayed planting. The 97 CRM DroughtGard hybrid DKC 47-27, 110 CRM DroughtGard DKC 62-27, and 109 CRM conventional hybrid DKC 59-92 produced 0.3 to 0.7 more ears m^{-2} in 2014 than 2013, while the 100 CRM DroughtGard hybrid DKC 50-57 produced 0.5 fewer ears m^{-2} in 2014 than 2013. This variable response was unexpected since the number of ears m^{-2} is determined by the V6 growth stage (Abendroth et al., 2011), and because the early season monthly rainfall in May was 25 mm greater in 2013 than in 2014 (Table 2). Swanson and Wilhelm (1996) found no difference in the number of ears m^{-2} across planting dates, and Nemali et al. (2015) found similar number of ears m^{-2} for DroughtGard and conventional maize hybrids in well-watered environments.

In general, the number of kernels ear^{-1} and kernels row^{-1} (data not presented) increased with delayed planting, with the exception of the number of kernels ear^{-1} for the 20 May planting

Table 4. Influence of year by hybrid on maize grain yield and yield components.

Hybrid	Grain yield	Rows ear^{-1}		Kernel depth		Ear circumference		Kernel weight		Bulk density	
		2013	2014	2013	2014	2013	2014	2013	2014	2013	2014
	$Mg\ ha^{-1}$	no.		mm		cm		g 100 kernels $^{-1}$		kg m^3	
DKC 47-27	10.9B†	15.4Ea‡	15.8Da	11.8Ca	12.8Ba	14.1Ca	14.6Ca	31.0Aa	33.3Ba	75.8Aa	76.1Ba
DKC 50-57	11.2B	16.5Da	16.6Ca	13.2Ab	13.7Ab	14.8Bb	15.7Ba	30.5Ab	35.1Ba	72.2Eb	73.2Ca
DKC 62-27	13.0A	18.1Ba	18.6Aa	12.6ABa	13.1Ba	14.8Ba	15.3Ba	30.1Aa	31.2Ca	74.2Cb	77.2Aa
Channel 214	13.0A	17.2Ca	17.5Ba	12.0Cb	13.9Aa	14.6Bb	16.6Aa	27.0Bb	39.8Aa	73.2Db	76.8ABa
DKC 59-92	13.0A	19.0Aa	18.3Aa	12.5Bb	13.2Ba	15.0Ab	15.7Ba	25.8Bb	30.9Ca	74.5Ba	76.4ABa
Mean	12.2	17.2a	17.4a	12.6b	13.3a	14.7b	15.6a	28.9b	34.1a	74.0b	75.9a

† Means followed by the same capital letter in each column were not significantly different at $P \leq 0.05$.‡ Means followed by the same small letter in each row for year of each parameter were not significantly different at $P \leq 0.05$.**Table 5.** Influence of hybrid by planting date within year on number of maize ears m^{-2} .

Hybrid	2013				2014			
	27 Apr.	16 May	3 June	Mean	18 Apr.	5 May	20 May	Mean
	Ears m^2							
DKC 47-27	5.5Ba†‡	5.6ABb	5.9Aa	5.7A	6.1ABa	5.9ABCa	6.0Aa	6.0B
DKC 50-57	5.7Aa	5.8ABa	5.9Aa	5.8A	6.2ABa	5.1Bb	4.5Cc	5.3C
DKC 62-27	5.3Bb	5.2Bb	5.6Ba	5.4B	6.2ABb	7.7Aa	5.9Ab	6.6A
Channel 214	5.2Ba	5.6ABa	5.6Aa	5.5B	5.5Ba	5.7Ca	5.1Bb	5.4C
DKC 59-92	5.8Ab	6.0Aa	6.0Aa	5.9A	6.7Aa	6.2Bab	5.7Ac	6.2B
Mean	5.5a	5.6a	5.8a		6.1a	6.1a	5.4b	

† Means followed by same capital letter in each column were not significantly different at $P \leq 0.05$.‡ Means followed by the same small letter in each row for year of each parameter were not significantly different at $P \leq 0.05$.

date in 2014 (Table 6). This result contrasts with finding of Swanson and Wilhelm (1996) who found production of the most kernels for mid-May planting, and Lindsey and Thomison (2016) who found reduced production of kernels with delayed planting. The conventional hybrid DKC 59-92 produced the greatest number of kernels ear⁻¹ and kernels row⁻¹ for planting dates between 18 April and 20 May, while the 112 to 114 CRM hybrids DKC 62-27 and Channel 214-00 (hereafter referred to as Channel 214) produced an intermediate number, and the 97 to 100 CRM hybrids produced the lowest number of kernels ear⁻¹ (Table 6). Considerable variation among hybrids occurred for the later planting dates of 3 June in 2013 and 20 May in 2014, but averaged over planting dates, the DKC 59-92 conventional hybrid produced more kernels ear⁻¹ than DroughtGard hybrids. Nemali et al. (2015) indicated that DroughtGard hybrids produce more kernels ear⁻¹ than conventional hybrids in water-limited environment and a similar number of kernels ear⁻¹ in well-watered environments.

Ear length was greater in 2014 than 2013 (Tables 3 and 6), but was not consistently influenced by the planting date (Table 6). DKC 62-27 and Channel 214 produced the greatest ear length in 2013 and Channel 214 in 2014. Although statistically significant, only small differences in ear length were present in both years.

Analysis of variance indicated that the planting date within year main effect influenced the kernel depth, kernel weight, and bulk density (Table 3). Kernel depth, kernel weight, and bulk density were greater in 2014 than in 2013 (Table 4). In 2013, delayed planting from 27 April to 16 May reduced kernel depth and kernel weight with no further reduction as planting was delayed to 3 June (Table 7). In 2014 with higher seasonal rainfall, delayed planting from 18 April to 20 May increased the kernel depth, kernel weight, and bulk density. These results contrast with results of Kaiser et al. (2016), Lindsey and Thomison (2016), Bonelli et al. (2016), and Van Roekel and Coulter (2011) who found that kernel weight decreased with delayed planting date, but are consistent with Swanson and Wilhelm (1996) that found that planting date had a lower magnitude of influence on kernel weight than other yield components. Early maturity hybrid DKC 47-27 had lighter kernel weight than other hybrids, while DKC 50-57 had the greatest kernel depth (Table 4). Kernel weights among hybrids in both years was quite variable (Tables 4 and 7). DKC 47-27 produced greater bulk density in 2013, and DKC 50-57 had the lightest bulk density in both years.

Analysis of variance indicated that the number of rows ear⁻¹, kernel depth, ear length, ear circumference, kernel weight and bulk density were influenced by the Year × Hybrid interaction effect (Table 3). In 2013, kernel depth, kernel weight, and bulk density decreased with delayed planting date while the opposite occurred in 2014 (Table 7). The number of rows ear⁻¹, kernel depth, ear length, ear circumference, kernel weight, and bulk density were greater in 2014 than in 2013 (Table 4) corresponding to greater rainfall in April and May 2013 and more rainfall in June through September in 2014 (Table 2). The conventional hybrid DKC 59-92 had the greatest number of rows ear⁻¹ (along with DKC 62-27 in 2014) (Table 4) and one of the shortest ear lengths (Table 6) in both years, while the 97 CRM hybrid had the fewest number of rows ear⁻¹, kernel depth and ear circumference in both years (Tables 4

Table 6. Influence of hybrid by planting date within year on number of maize kernels ear⁻¹ and ear length.

Hybrid	Kernels ear ⁻¹						Ear length					
	2013			2014			2013			2014		
	27 Apr.	16 May	3 June	Mean	18 Apr.	5 May	27 Apr.	16 May	3 June	Mean	18 Apr.	5 May
DKC 47-27	513Ba†‡	545Da	556Ca	538C	602ca	597ca	16.6Ba	17.3Aab	18.1Aa	17.3AB	19.7Aa	19.3Ba
DKC 50-57	540Ba	523Ca	624Ba	562C	609ca	629bda	17.1Ba	15.9Ab	18.2Aa	17.1AB	19.5Aa	19.5Ba
DKC 62-27	633Aa	659Aa	599Ba	630B	694ABab	728Ba	18.0Aa	18.7Aa	17.1Ab	17.9A	18.5Aa	19.4Ba
Channel 214	580Aba	615Bab	642Ba	612B	675Ba	684Ba	18.2Aa	17.1Aa	17.8Aa	17.7AB	21.3Aa	20.7Aa
DKC 59-92	663Aa	705Aa	716AA	695A	751Aab	806Aa	16.2Bb	18.1Aa	16.2Aa	16.8B	18.9Aa	20.1ABa
Mean	586a	609a	627a	666b	666b	689a	17.2a	17.4a	17.5a	17.5a	19.6a	19.8a

† Means followed by the same capital letter in each column were not significantly different at $P \leq 0.05$.

‡ Means followed by the same small letter in each row for year of each parameter were not significantly different at $P \leq 0.05$.

Table 7. Influence of planting date within year on grain yield components.

Planting date	Kernel depth mm	Kernel weight mg 100 kernels ⁻¹	Bulk density kg m ³
2013			
27 Apr.	12.9A†	31.5A	74.7A
16 May	12.2B	27.8B	74.4A
3 June	12.2B	27.4B	72.9B
Mean	12.4b‡	28.9b	73.9B
2014			
18 Apr.	13.0C	32.7B	74.9B
5 May	13.4B	32.7B	76.2A
20 May	13.6A	36.7A	76.8A
Mean	13.3a	34.0a	76.0A

† Date means followed by the same capital letter in each column within each year were not significantly different at $P \leq 0.05$.

‡ Means followed by the same small letter across years were not significantly different at $P \leq 0.05$.

and 6) and the 114 CRM hybrid Channel 214 had the greatest ear length, ear circumference, kernel weight and bulk density in 2014 (Tables 4 and 6) with the higher seasonal rainfall year than 2013 (Table 2). In 2013, the 97 CRM hybrid DKC 47-27 had the greatest bulk density, and in 2014, the 109 to 114 d CRM hybrids DKC 62-27, DKC 59-92, and Channel 214 had the greatest bulk densities (Table 4). Nemali et al. (2015) found inconsistent kernel weight differences between DroughtGard and conventional maize hybrids.

Pearson Correlations

Maize grain yield was positively associated with yield components measured in this study, but yield was similar across planting date while yield components varied (Table 8). Yield components associated with grain yield for the early planting date were number of ears m⁻², rows ear⁻¹, kernels ear⁻¹, kernels row⁻¹, ear length, and ear circumference, while grain yield was associated with kernel depth, ear circumference, and kernel weight for the late planting date. All yield components were correlated with grain yield for the intermediate planting date. For early planting dates the yield components of number

of kernels ear⁻¹ and kernels row⁻¹ were associated with the number of ears m⁻², while late planting date was associated with kernel depth, ear length, ear circumference, and kernel weight. Relations between the number of kernels ear⁻¹ with late occurring yield components of kernel depth, ear length, ear circumference and kernel weight were less associated at late planting dates, while late-planted corn kernel weight was associated with early season yield components of number of ears m⁻², rows ear⁻¹, and kernels row⁻¹. The grain yield and yield component data suggest that relatively early developing yield components of ears m⁻², number of kernels ear⁻¹, and kernels row⁻¹ (Abendroth et al., 2011) were of most importance for early planted maize while late-developing yield components of kernel depth and weight were of most importance for late-planted maize. Since seasonal rainfall was near average in both years, except in July which was lower, suggests that yield components were most influenced by early or late-season stresses, which varied with planting date and were associated with grain yield compensation to produce similar yield.

CONCLUSION

Planting date had no influence on grain yield but different planting dates led to compensation among yield components to maintain grain yield, confirming that yield component compensation is associated with variable response to planting date as well as with plant population (Milander et al., 2016; Reeves and Cox, 2013). Maize grain yield was associated with all yield components, but variation occurred among planting dates. Early developing yield components were relatively more important for earlier planted maize while late-developing yield components were relatively more important for late-planted maize. No differences in maize grain yield were found between DroughtGard and conventional hybrids with the same maturity classification, while late-maturity hybrids produced higher grain yields than early maturity hybrids. These data indicate that planting date and drought-type hybrid in rotation with soybean had little effect on maize grain yield under the relatively high rainfed conditions present in this Nebraska study due to grain yield component compensation.

Table 8. Pearson correlations and probability of maize grain yield and yield components at three different planting dates, averaged over 2013 and 2014.

Primary yield component	Planting date	Ears m ⁻²	Rows ear ⁻¹	Kernels ear ⁻¹	Kernels row ⁻¹	Kernel depth	Ear length	Ear circumference	Kernel weight	Bulk density
Yield	1	0.58**	0.43*	0.67**	0.60**	0.23	0.60**	0.45*	0.09	0.18
	2	0.49**	0.55**	0.72**	0.71**	0.62**	0.73**	0.71**	0.40*	0.51**
	3	0.10	0.27	0.33	0.13	0.48**	0.26	0.60**	0.45**	0.66**
Ears m ⁻²	1		0.22	0.41*	0.43*	0.06	0.18	0.04	-0.15	0.11
	2		0.51**	0.42**	0.28	0.13	0.16	0.14	-0.11	0.47**
	3		0.14	-0.07	-0.33	-0.42*	-0.41*	-0.48**	-0.41*	-0.02
Kernels ear ⁻¹	1	0.41*	0.78**		0.81**	0.30	0.48**	0.57**	-0.04	0.35
	2	0.42*	0.83**		0.86**	0.41*	0.67**	0.63**	0.08	0.45*
	3	-0.07	0.53**		0.43*	0.26	0.10	0.36	-0.07	0.23**
Kernel weight	1	-0.15	-0.35	-0.04	0.17	0.66**	0.62**	0.69**		-0.08
	2	-0.12	-0.16	0.08	0.35	0.65**	0.65**	0.70**		0.40*
	3	-0.41*	-0.39*	-0.07	0.49**	0.82**	0.80**	0.80**		0.67**

* Significant at the $P \leq 0.05$.

** Significant at the $P \leq 0.01$.

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